

PROFESSIONAL SUMMARY

BTech Computer Science student and software developer skilled in Python, C++, Flutter, Dart, JavaScript, and AI/ML. Built practical projects including AgriSmart AI, student support applications, and university examination automation tools. Experienced in web development, REST APIs, databases, machine learning, and cross-platform applications, with familiarity in FastAPI, React, Firebase, SQL, Git/GitHub, Docker, and Kubernetes. Passionate about AI-powered solutions, automation, and developing practical real-world software.

PROJECTS

AgriSmart AI — Intelligent Smart Agriculture Platform | *Python, FastAPI, Machine Learning, React* | Aug 2026

- Developed an AI-powered agriculture platform providing crop recommendations, disease detection, weather insights, and agricultural guidance.
- Integrated machine learning, real-time weather data, and an AI agriculture chatbot to support data-driven farming decisions.
- Built a modular architecture with REST APIs and database integration for reliable data management.

InteractED — Interactive CS Visualization App | *Flutter, Dart, Material 3, AnimationController* | Nov 2025

- Co-built a cross-platform Android & iOS educational app featuring interactive CS simulations including BFS, DFS, OSI, Virtual Memory/TLB, and ARP Spoofing.
- Engineered smooth 60 FPS animations using AnimationController, Tween, and custom Canvas painters.
- Developed the application using a single Flutter codebase for cross-platform deployment.

Akal Student Support App | *Flutter, Firebase, Python* | 2026

- Developing a real-time student support application using Firebase Firestore and secure multi-role authentication.
- Implementing Python-based automated report generation and data-processing workflows.
- Designing features to streamline student queries and university support services.

Exam Seating Plan Generator | *Python* | May 2026

- Developed an automated system to generate examination seating arrangements for Akal University, reducing manual effort.
- Implemented efficient student and room allocation logic for fast and reliable scheduling.
- Adopted by the institution and received an appreciation letter from Akal University.

Datesheet Generator | *Python* | 2026

- Developed a Python-based system to automate university examination datesheet generation.
- Implemented scheduling logic for organizing subjects, dates, and examination slots efficiently.
- Replaced manual scheduling with a faster and more reliable automated workflow.

ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

- Co-Lead Developer** – Akal Student Support App (end-to-end development)
- Member** – Ink & Pixel Club (2024–Present)
- Member** – Web Crafters Club (2024–Present)
- Appreciation Letter** – Awarded by Akal University for the Exam Seating Plan Generator project

EDUCATION

Expected in 05/2027

Akal University, Talwandi Sabo, Punjab

Bachelor of Technology: Artificial Intelligence & Machine Learning

- Pursuing (Undergraduate)
- Current SGPA: **8.24**

SKILLS

- Languages:** C/C++, Python, SQL, HTML, CSS, JavaScript (Basic), Dart
- Mobile/App:** Flutter, Android Studio, Firebase Auth, Cloud Firestore, Realtime Database
- AI & ML:** Machine Learning, Deep Learning, NLP (Basic), Reinforcement Learning, AI Agents, Prompt Engineering
- Data Tools:** Pandas, NumPy, Scikit-learn, TensorFlow (Basic), PyTorch (Basic)
- Dev Tools:** Gradio, Git, GitHub, Docker (Basic), Jupyter, Google Colab
- Systems:** Hardware & software installation, Web-based software engineering, Virtualization

CERTIFICATIONS

- Agentic AI: From Learner to Builder** (AI Agent Architect), IBM SkillsBuild & CSRB0X
- Introduction to Cloud Computing** (Aug 2026)
- Agentic AI: From Learner to Builder** (Aug 2025)
- The Complete Full-Stack Web Development Bootcamp** (Jul 2025)

KEY COURSES TAKEN

- Data Structures & Algorithms
- Database Management Systems
- Operating Systems
- Computer Networks
- Machine Learning & Deep Learning
- Computer Vision (Basic)